

Questions Used in the Study: Developing a Roadmap to an AI-first Organization: A Case Study in Embedded Software Development

No.	Question	Type	Options / Items
1	If you were to compare generative AI in SE to something, what would it be?	Choices	(a) Introduction of a compiler (b) Introduction of WWW (c) Introduction of Cloud computing (d) Introduction of Agile (e) Introduction of Git and CI/CD
2	If your AI assistant (Claude Code) worked on your project for one week, what would you need to do after?	Ranking	(a) Check compliance with requirements (b) Check compliance with our coding guidelines (c) Check compliance with architecture (d) Run all static analysis tools (e) Run all dynamic/runtime profiling tools (f) I would not do anything; I trust my AI
3	If you encountered a defect in AI-generated code, what do you do?	Choices	(a) I change the prompt in order to fix the bug (b) I edit the file myself (c) I change the <code>agent.md</code> file to avoid this in the future (d) I create a bug report hoping that someone else will fix it (e) I try a different AI agent
4	How would you like to see AI in your team?	Choices	(a) No AI, only humans (b) Humans who use AI agents (c) AI agents work and ask humans for help/directions (d) AI agents work and humans only direct them (e) No humans, only AI
5	Which technology will have the most impact?	Prioritisation	(a) Agentic AI (SE 4.0) (b) Programming assistants (like Copilot) (c) Code review assistants (d) Prompt engineering (Software 3.0) (e) Intelligent test assistants (f) CI/CD assistants
6	Which roles will need to adapt the most (re-skilling)?	Prioritisation	(a) Programmers (b) Architects and designers (c) UX designers (d) Testers (e) Requirements engineers (f) Release engineers (g) Project managers
7	How important is it to build the competence in-house for the following?	Prioritisation	(a) Developing agents (b) Prompt engineering (c) AI engineering (d) Integrating AI into products (e) Using AI for SE (f) Software-on-demand
8	What hinders the adoption of these tools?	Open-ended	<i>Open-ended response</i>
9	If you were to invest in improving AI, what would you improve?	Prioritisation	(a) Speed (b) Hallucinations (c) Termination criteria (when to stop) (d) Time frame for solving tasks (e) Data leakage (from prompts/tools) (f) Transparency in generation (g) Oracles for validation
10	If you were to invest in your organization, what would it be?	Prioritisation	(a) AI-friendly code (b) New competence (c) Dedicated AI team (d) New models, better models (e) New tools, better tools (f) Integration with current tools (g) Refactoring/rewriting/redesigning legacy software
11	What will happen to your team in the future (3–5 years)? Scales <i>Context: The question and related discussion were about which aspects of each team would be affected and which response options they think they will see more or less of. Scale: Much less, Less, Unchanged, More, Much more.</i>	Scales	(a) Agentic AI (b) AI support (c) Cross-functionality (d) Number of people (e) Interaction with external stakeholders (f) Typical SE roles (g) Programming done by non-programmers
12	What will happen to your organization (3–5 years)? Scales <i>Context: The question and related discussion were about how much the need for each part of the organization will change in the future. Scale: Much less, Less, Unchanged, More, Much more.</i>	Scales	(a) SE roles (b) AI roles (c) Sales roles (d) Product management (e) Administration, e.g., HR, managers (f) Domain specialists (g) Non-SE roles do SW development, e.g., vibe coding (h) New products/diverse products
13	What will happen to software engineering (3–5 years)? Scales <i>Context: The question and related discussion were about how much the need for each role will change in the future. Scale: Much less, Less, Unchanged, More, Much more.</i>	Scales	(a) Programmers (b) Project managers (c) Requirements engineers (d) UX designers (e) Architects/designers (f) Testers (g) Product managers (h) Process owners/managers

Types of questions: **Choices** - Choose one option. **Ranking** - Rank the options in order from most to least agreed with. **Prioritization** - Give each option a score between 0-100. **Scale** - Rate each option on a scale from -2 to 2.